Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/21/2016
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Emily (NASA Glenn Research Center) Sheehan, Christopher (ZIN Technologies, Inc.) Loewen, Hartmut (University of Dusseldorf) Telo da Gama , Margarida Ph.D. (University of Lisbon) Yethiraj, Anand (Memorial University of Newfoundland)		
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Task Description:	Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We will use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and planned in the summer of 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
	Major Accomplishments The study was conducted within the scope of the originally-proposed research plan. The New Jersey Institute of Technology (NJIT) team worked in closed collaboration with NASA scientists on preparing International Space Station (ISS) Advanced Colloids Experiment (ACE) experiments. The main focus was on the development of requirements for a high-voltage amplifier and an optical system. The NJIT team carried out ground-based experiments on neutrally buoyant particles to specify basic parameters for the design of the dielectrophoretic cell geometry, electrode configuration, voltage/frequency regimes, and preparing suspensions needed for the ISS experiments. It was shown that new essentially periodic structures and patterns are formed in the presence of electric fields. Moreover, both AC and DC are essential to the formation of new structures. On October 19 - 20, 2015, the Principal Investigator (PI) participated in the ACE-E Science Concept Review (SCR) of the proposed ISS ACE-E experiments at NASA Glenn Research Center. The SCR Committee confirmed that long-duration microgravity is necessary for this project due to a large density difference between metallic (aluminum) particles and suspending oil and supported the proposition to fly experiments for developing a deeper understanding of non-equilibrium colloidal assembly in temporally and spatially varying electric fields. The SCR Committee reaffirmed: • The scientific and technical merit and significance of the research objectives. • The need for a long-duration microgravity environment. • The proposed ACE-E2 experiments are feasible and the hardware selection is reasonable. • The selection of the test matrix is adequate. • The proposed selection of samples for the test matrix spans the regions that have shown interesting phenomena in ground experiments. • The idea of using low frequency fields at 0.5 Hz to stir and randomize a sample after exposure to an electric field is excellent. Following SCR Committee’s recommendations, NJIT team demonstrated: • Cargill dielectric fluids Envirotemp E200 and Envirotemp FR3 can be used for replacing corn/mazola oil in ISS experiments. These fluids are stable, well characterized and a particle structure patterns formed in a suspension is similar to that observed in corn/mazola oil.

Task Progress:	• A cellular structure formed in an experiment conducted after electro-hydrodynamic stirring of a cellular structure formed in the previous experiment does not take the same spots. Specifically, it was shown that separations between particle-free domain centers in both cellular structures follow the Gaussian distribution. Development of Requirements for ISS experiments Hardware: NJIT team will use the LMM Leica RXA microscope with low power objectives (1X and 2.5X) to study the electric field and temperature effects in colloids in cells 0.4 to 1 mm thick and at least 20 times larger in the other two directions. A setup to be developed for experiments in the ISS should provide AC high-voltage to energize electrodes of a dielectrophoretic cell and control the sample temperature. It is proposed to study samples with at least two different gaps and study each sample at three to six sets of AC electric field and frequency and DC offset. It is desirable to conduct several tests at each voltage and at least two temperatures. There will be a need to make several cells since a bubble formed in a sample in the ISS, even when degassed in preparation in the Earth facility, makes the sample cell unusable. Ground based experiments conducted by NJIT team demonstrated that requirements for the setup can be simplified: • A field frequency range in ISS experiments can be reduced from MHz to 1-2 kHz. • Placing one of the ITO electrodes outside of the dielectrophoretic cell should extend lifetime of a sample. Measurement of suspension conductivities before and after pattern formation/stirring can be used to ensure that the ITO coated surface does not dissolve in the suspension. Materials: NJIT team conducted ground-based experiments to specify the particle type, the suspending liquid, and the number and volume of specimens of different concentrations within each type. We expect to conduct ISS experiments on several types of particles (aluminum, polyethylene, or polyalphaolefin) and use several concentrations for each type of particles to cover a wide range of the mismatch of dielectric constants and conductivities between particles and a suspending liquid. Nearly buoyancy-matched suspensions of polyethylene or polyalphaolefin would serve as “control samples” for comparing results of ground-based and microgravity tests. Operational requirements: A dielectrophoretic cell for ISS experiments will be fully filled with a suspension in the Earth facility. As the particles will sediment in Earth’s gravity during storage and the hyper-gravity during the spacecraft launch, specimens should be homogenized prior to experiments. A homogenized sample will be heated to a desired temperature and then exposed to an electrical field by applying a desired voltage/frequency for a certain period of time. To quantify effects of electric fields on colloidal processes, experiments will be carried out on each sample at different voltages and frequencies and, if possible at several different temperatures. After exposure to an electric field, a sample should be homogenized prior to another test. For each suspension, 3 - 6 magnitudes of applied voltage and several frequencies at each voltage will be tested for each selected temperature. Several tests at selected voltage/frequency regimes will be repeated to evaluate reproducibility of microgravity experiments. The voltage/frequency regime and duration of a particular microgravity experiment will be determined in real-time depending on the previous results as data sets are transmitted to Earth. These parameters cannot be predicted based on observations of the field-driven processes in non-buoyancy-matched suspensions under Earth’s gravity. NJIT team demonstrated that it is possible to homogenize a sample by employing an electro-hydrodynamic flow generated in the dielectrophoretic cell by a low frequency electric field. It would allow conducting several microgravity tests on each sample. This technique was successfully tested in ground-based experiments. Planned Work for Next Year 1. Continue working with NASA project scientists on the selection of an amplifier, voltage/frequency regimes, dielectrophoretic cell design, and suspensions for ISS tests. 2. Continue ground-based experiments needed to specify science requirements for experiments aboard the ISS.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Khusid B, Elele EO, Shen Y, Pettit DR. "How does electricity make liquid bristle?" Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Concurrent Session 21: Fluids Physics #4, November 13, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Elele EO, Shen Y, Pettit DR, Khusid B. "Electric field control of mixing and separation processes inside a tiny droplet." Presented at the AIChE 2015 Annual Meeting, Salt Lake City, UT, November 9, 2015. AIChE 2015 Annual Meeting, Salt Lake City, UT, November 9, 2015. Abstract 425347, https://aiche.confex.com/aiche/2015/01/papers/viewonly.cgi?password=949217&username=425347 , Nov-2015
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